

Design review package (r4.11): fewer seams, fewer invariants

- **For:** reviewers (people or models). This document stands alone; every file it names is in spark-daemon-pattern/ on branch claude/clever-bardeen-qkqxez.
- **The ask:** find design simplifications that **remove a seam or an invariant without weakening any guarantee in §3**. We would rather have one elegant removal than ten new checks.
- **Status:** FOR REVIEW. Nothing here is adopted. The candidates in §5 (E-1 to E-9) are ours, offered to be beaten.
- **Revision:** r4.11, 2026-10-06, by Claude. Code candidate 926720f; contract 4.0.0 (dfdb9f1b...).
- **Evidence:** 267 of 267 self-tests, none skipped; the four reference daemons pass all 22 battery checks (evidence/r4.11/). The rulings as built are recorded in ADJUDICATION-R4.9-OWNER-IMPLEMENTATION.md.

How to mark: for each seam (§4) and candidate (§5), say KEEP, SIMPLIFY (how) or REMOVE, and name the evidence type: reproduced, measured, traced to code, or opinion. Then add your own candidates. A good candidate names:

- the seam or invariant it removes;
- the guarantee in §3 that still holds without it, and why;
- what would make it wrong.

1. What r4.11 built, and what it taught

The owner's rulings on R-1 to R-8 are built, except R-5c, which is gated because no checkpoint anchor exists. In one line each:

Ruling	Built as
R-7	One judge (judge.py): validate, precheck and battery are profiles of one check registry with one verdict rule
R-3	An installable unit comes only from a qualifying battery PASS, with a qualification record. Two gates: qualified for the installer, and the runtime's digest check
R-1	The manifest's deny list is gone; no read may contain an always-denied path
R-2b	Execute only on declared executables and their loaders. A command-free daemon can run nothing. One residual: the loader can be run on another readable binary
R-6	cycle_budget_seconds: one deadline per cycle, enforced by ctx timeouts and an alarm; unit timings are derived from it
R-4	One interpreter (semantics.py) for start-up and status; two independent verifiers, compared by DB-22
R-8b	A rename map and a three-hash binding for vendored copies

Three defects were found this round. Each sat on a seam, and each says something about simplifying:

Defect	The seam it sat on	What it suggests
HF-38 (Low): a record with "seq": true verified as seq 1 (True == 1 in Python)	Between the writer and its readers. Both verifiers had the blind spot until the second one was written	Diversity found it. Keep verification diverse, and keep interpretation single (E-8)
HF-39 (Low): the battery report named its rewritten workspace copy of the manifest, not the manifest that would run	Between the manifest as written and the battery's copy (the same seam as HF-37 in r4.7)	Two copies of one identity drift. Remove the copy rather than guard it (E-3)

Defect	The seam it sat on	What it suggests
HF-40 (Low): a command could outlive its cycle, because the alarm fired inside <code>proc.run</code> 's cleanup	Between two enforcers of one budget: <code>proc</code> 's timeout and the cycle alarm	Two enforcers of one rule create a seam between them. Choose one (E-2)

2. The design today, by count

What	Count	Notes
Package code	6,115 lines in 24 modules (plus 199 in the independent verifier and 131 in the vendoring tool)	The largest are <code>battery.py</code> 799, <code>runtime.py</code> 584, <code>manifest.py</code> 494, <code>contract.py</code> 400
Schemas a reader may meet	12	<code>ledger/1</code> , <code>manifest/3</code> , <code>contract/1</code> , <code>candidate/1</code> , <code>validate/2</code> , <code>precheck/2</code> , <code>battery/2</code> , <code>qualification/1</code> , <code>status/1</code> , <code>independent-verifier/1</code> , <code>rename-map/1</code> , <code>vendor-binding/1</code>
CLI commands	15	12 public (<code>describe</code> , <code>schema</code> , <code>scaffold</code> , <code>envelope</code> , <code>validate</code> , <code>precheck</code> , <code>unit</code> , <code>run</code> , <code>verify</code> , <code>status</code> , <code>battery</code> , <code>qualified</code>) and 3 internal probes
Battery checks	22	DB-01 to DB-18, DB-20, DB-22, DB-24, DB-25 (DB-19, 21, 23 reserved)
Seams named for review	11	\$4
Invariants a reviewer must hold the design to	8 owner invariants, plus 9 core invariants proposed in r4.9	\$3; they overlap (E-8)
Behaviours test mode relaxes	7	Poll interval, blind limit, cycle budget, audit mode, jitter, tolerance of missing Landlock, and unqualified starts; refused under <code>systemd</code> (E-6)
Self-tests	267	

3. Guarantees that must not weaken

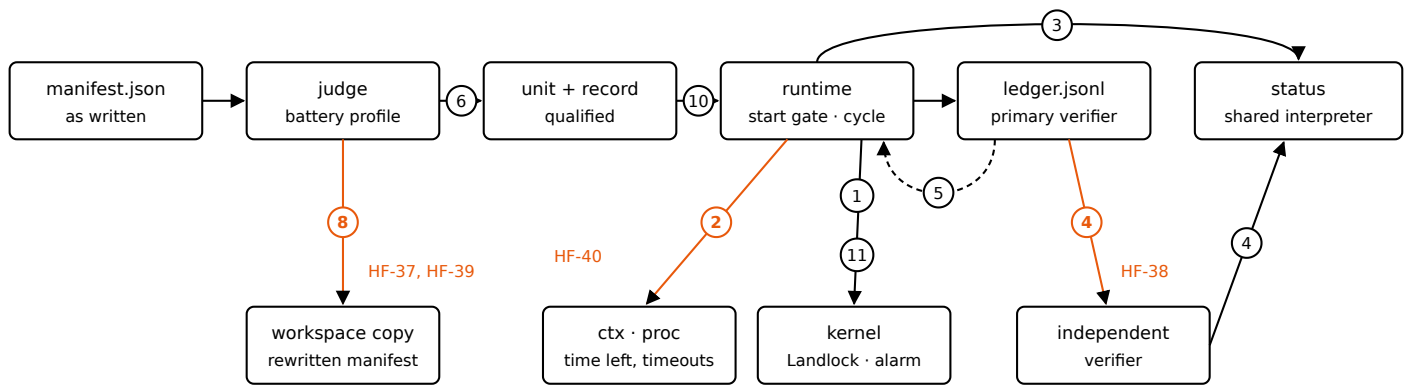
These are the fixed points. A simplification that weakens one is not a simplification.

1. A battery PASS is qualification evidence, not permission to activate.
2. No daemon class above `observe` is unlocked.
3. A command declaration is an allowlist, not a Boolean switch.
4. Start-up and status share interpretation; corruption and tamper detection stays diverse.
5. Integrity uncertainty is always visible, never rendered as healthy.
6. `cycle_budget_seconds` constrains the whole cycle.
7. Activation is a separate authority decision after qualification.
8. Nothing relaxes confinement, blindness, authentication, or proposal-versus-authority.

The owner's rulings that bound the design also stand:

- **PD-63:** the blind period; exit 78, never restarted.
- **PD-70:** blindness survives restarts.
- **PD-01.9:** the Blind Forester, outside observe-only daemons.
- **Break Glass:** never for internal systems, and "an AI model cannot unilaterally decide that an emergency exists".

4. The seams in r4.11



Where the seams sit in r4.11. Numbers match the table below; orange marks the three where this round's defects were found. Seam 3 is the shared interpreter (start-up and status), 4 the two verifiers feeding it, and 5 the checkpoint R-5c would add (dashed: not built). Seams 7 (vendoring) and 9 (test mode) are not drawn.

#	Seam	What each side assumes	What enforces it today	Found this round
1	Manifest commands → kernel execute grant	Every declared command resolves to one ELF file with one loader	landlock.execute_paths; test_landlock.py	The loader residual (R-2b)
2	Cycle budget → ctx/proc timeouts → alarm → watchdog → stop timeout	Exactly one thing ends an overrunning cycle, cleanly	context._remaining, the runtime alarm, proc's timeout; derived unit timings; test_budget.py	HF-40 (the two enforcers met)
3	Shared interpreter → start-up and status	Both feed it the same records and clock readings	semantics.py; test_independent.py	—
4	Two verifiers → interpreter	Only an intact chain is interpreted; the verifiers agree	Verified.from_facts; DB-22	HF-38 (a shared blind spot)
5	Full verifier → checkpoint → start-up	(not built: no anchor)	Full start-up verification	—
6	Battery result → unit generator	The unit judged is the unit emitted	Unit options are battery inputs; DB-15 and DB-25	—
7	Source tree → rename map → copy	The copy is the source under the map	vendor.py --check	—
8	Manifest as written ↔ battery workspace copy	Both name one identity	Reports bind the original; run checks use the copy	HF-39 (and HF-37 in r4.7)
9	Test mode ↔ production	The seven relaxations never reach a service	INVOCATION_ID refusal; overrides recorded in DAEMON_START	—
10	Qualification record ↔ unit ↔ runtime gate	Two gates check overlapping things: unit bytes and host (installer), code and contract (runtime)	qualify.check; the runtime's digest check	—
11	Python deny checks ↔ Landlock ↔ the unit's InaccessiblePaths=	They express one policy	Validation (no gaps) makes them agree on paths as written; a symlink created later is caught by Python (exit 78) and by Landlock (EACCES), with different outcomes	—

5. Candidate simplifications

Ordered by what they would remove. Each one is ours and unadopted, and each says what would make it wrong.

▼ E-1. One report shape, and no separate qualification record

Today:

- R-7 unified the judge, but its three profiles still write three report schemas (validate/2, precheck/2, battery/2).
- R-3 added a fourth artifact, the qualification record (qualification/1).

Proposal:

- One report schema for every profile, with `profile` and `qualifying` fields. They already exist.
- The battery report itself becomes the qualification evidence. It already binds the manifest and code digests, the contract, the unit options and the host facts.
- The installer gate then regenerates the unit from the report's inputs and compares bytes:
`qualified --unit U --report R.`

Removes: three schemas and one artifact, and narrows seam 10.

What would make it wrong: a report that must stay readable by consumers of the old shapes, or a reason to keep qualification evidence smaller than a report.

▼ E-2. One enforcer for the cycle budget

Today: HF-40 came from two enforcers of one deadline:

- `proc`'s own timeout, fed the time left by `ctx`;
- the cycle's alarm.

Option A (proposed): the alarm is the only deadline.

- `proc.run` takes no timeout. It guarantees, as it now does, that the process group dies on any exit path.
- `ctx._remaining` disappears.

Option B: keep both, as now.

Removes (A): half of seam 2, a parameter on every `ctx` call, and the race HF-40 sat in.

What would make A wrong: a blocking call the alarm cannot interrupt where a timeout could, for example a read in uninterruptible sleep. But neither mechanism helps there; the watchdog does.

▼ E-3. The battery never rewrites the manifest

Today: the battery copies the manifest and rewrites an absolute `output_dir` into its workspace, so two digests exist for one daemon. That seam has produced two defects (HF-37 and HF-39).

Proposal: run the battery's children with a test-mode output root instead. One override, recorded in `DAEMON_START` like the others, maps `output_dir` under the workspace. The manifest is never copied, and one digest is used everywhere.

Removes: seam 8 and its class of defect.

What would make it wrong: it adds a test-mode relaxation (see E-6), and a path mapping that a confused deployment could misuse. It is refused under systemd like every override.

▼ E-4. One outcome for a denied path

Today: validation forbids gaps, so for paths as written the Python deny checks, Landlock and `InaccessiblePaths=` agree. A symlink planted later inside a watched folder, pointing at a denied path, is refused by both Python and Landlock, but differently:

- **Python:** a policy violation, so the daemon stops with exit 78.
- **Landlock alone:** a failed read, which counts toward the blind limit.

Proposal: pick one outcome, then delete the code that only exists to produce the other.

- **If a failed read:** `guard.denied`, the audit hook's deny branch, `landlock.gaps()`, `DAEMON_START.landlock.gaps` and DB-18's gap comparison all go.
- **If a violation:** keep Python as the single detector, and drop `InaccessiblePaths=` for the always-denied paths.

Removes: seam 11, and one invariant (`gaps == []`).

What would make it wrong: whether a planted link is an attack signal that must stop the daemon (HF-17 argued that stopping is a denial of service by anyone who can write the folder).

▼ E-5. One gate, at start

Today: there are two gates.

- The installer's checks the unit bytes and the host.
- The runtime's checks the code, manifest and contract at every start.

Proposal: the runtime also checks the host facts. The qualified unit carries them, or a digest of them, in `ExecStart`. The installer's gate is then a convenience, not a guarantee.

Removes: the overlap in seam 10.

What would make it wrong: a kernel update would stop every daemon until it is qualified again. That may be exactly what G-5 asks for, or may be too strict for an owner.

▼ E-6. A smaller test mode

Today: one environment variable relaxes seven behaviours (§2).

Proposal: the battery runs the daemon qualified, with the digests it is judging, so "an unqualified start is allowed" is no longer a test-mode relaxation. The limit overrides become one recorded object instead of four variables.

Removes: part of seam 9.

What would make it wrong: the self-tests run many hand-made fixtures, and qualifying each one costs a digest computation per run (milliseconds).

▼ E-7. Decide R-2 by inventory

Today: one reference daemon (`git-watch`) runs a program, and the pilot's daemon runs none. The command surface carries:

- 11 of the 40 entries in HARDENING.md, HF-40 included;
- PD-93 and PD-94;
- half of seam 2 (E-2);
- seam 1 and the loader residual.

Proposal: count the daemons actually planned. If few need a program, adopt R-2: the core runs none, and commands become an extension that runs them through a broker outside the confined process (PD-72).

Removes: seam 1, the residual, proc.py from the core, and every command-related invariant.

What would make it wrong: if most planned daemons read Git through git, the commands belong in the core, and the broker is the work.

▼ E-8. Fewer invariants, each with one check

Today: the owner's 8 invariants (§3) and the 9 core invariants proposed in r4.9 overlap. For example:

- owner 3 and INV-1 and INV-2 are all about confinement;
- owner 4 and 5 and INV-5 and INV-6 are all about the ledger.

Proposal: six invariants, each enforced in one place and tested once:

#	Invariant	Absorbs
1	Observe only, and the kernel enforces it	Owner 2, 3 and 8 (confinement); INV-1, INV-2
2	Never record a guess	INV-3
3	Blindness surfaces within T	Owner 6; INV-4
4	The ledger is the one source of truth: written once, verified two ways, interpreted once, uncertainty visible	Owner 4 and 5; INV-5, INV-6
5	It runs only what was judged, and judging is not activation	Owner 1 and 7; INV-7
6	Bounded: one budget per cycle, limits derived, restart by class	INV-8, INV-9

What would make it wrong: an invariant that loses its own check when merged.

▼ E-9. Fewer commands

Today: verify (the primary verifier's head) and status (independent verification plus interpretation) overlap.

Proposal:

- verify becomes status --verify-only.
- scaffold and envelope are authoring tools that a frozen core need not carry.

Removes: one command and one reader path. A reader then always meets the same verification and interpretation.

What would make it wrong: tooling that already calls verify.

6. Questions for the reviewer

1. Which seam in §4 could be **removed** rather than guarded? Name the guarantee in §3 that survives without it.
2. Is there a second pair of enforcers of one rule, like HF-40's, that we have not found?
3. Is there another place where one identity has two copies, like HF-39's?
4. Of E-1 to E-9, which would you adopt first, and which would you reject? Why?
5. Is six invariants (E-8) the right number, or can two more merge?
6. What would you delete from the 6,115 lines without losing a guarantee?

7. Markup table

#	Item	KEEP / SIMPLIFY / REMOVE	Evidence type	Notes
1	Seam 1, command grant			
2	Seam 2, cycle budget (E-2)			
3	Seam 3, shared interpreter			
4	Seam 4, two verifiers			
5	Seam 5, checkpoint			
6	Seam 6, battery to unit			
7	Seam 7, vendoring			
8	Seam 8, workspace copy (E-3)			
9	Seam 9, test mode (E-6)			
10	Seam 10, two gates (E-1, E-5)			
11	Seam 11, deny outcome (E-4)			
12	E-7, R-2 by inventory			
13	E-8, six invariants			
14	E-9, fewer commands			
15	Your own candidate			